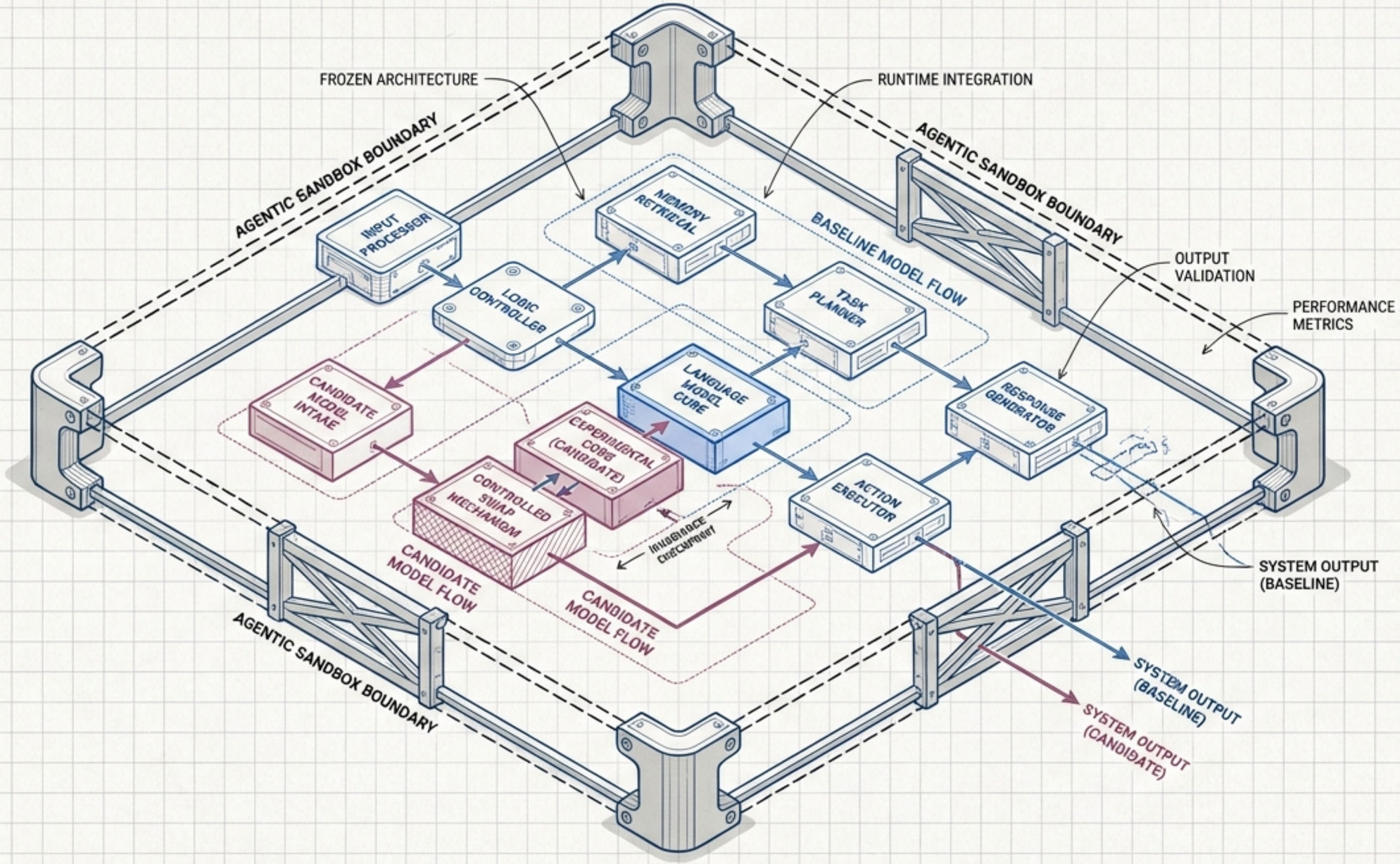
MIRA_ARCHITECTURE_V2 // EXECUTING_SANDBOX_BLUEPRINT

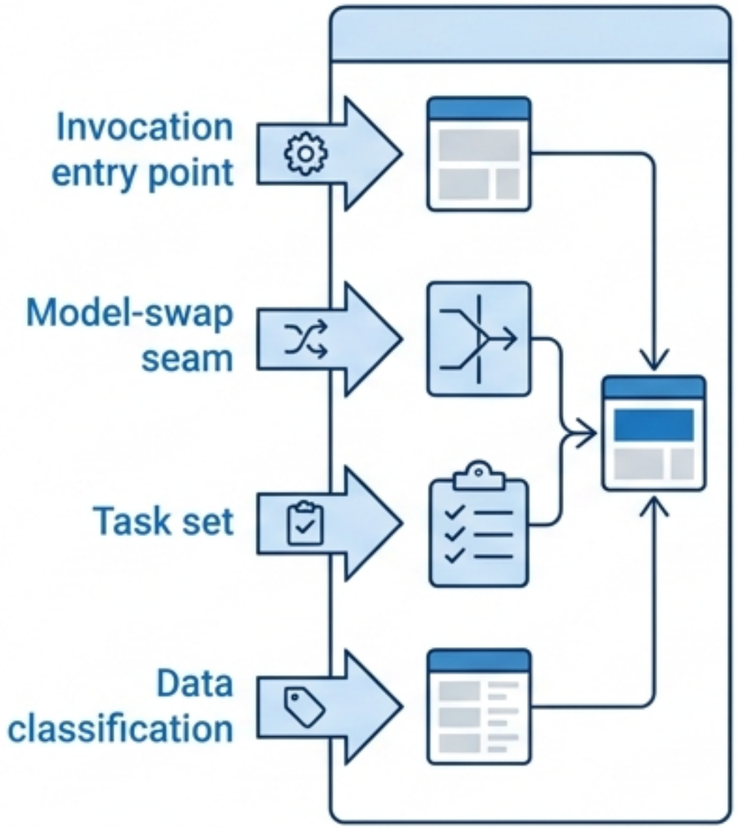
Model Invariance and Resilience Analysis for Agentic Applications



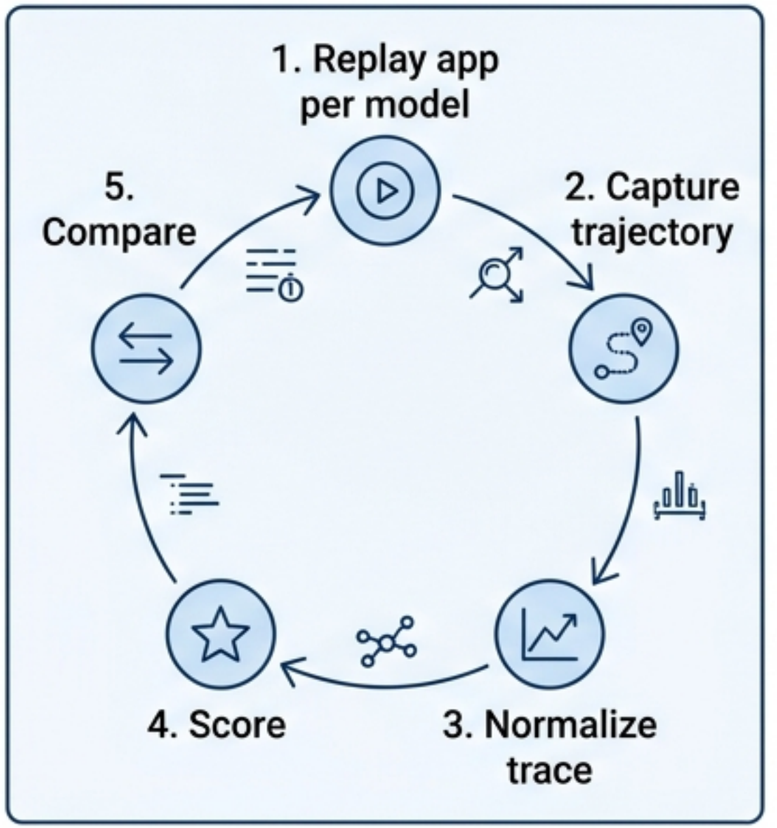
Gemini Notebook

A framework-agnostic clean room for evaluating model swaps

INPUTS

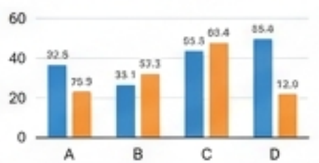


MIRA ENGINE



OUTPUTS

Per-model scores



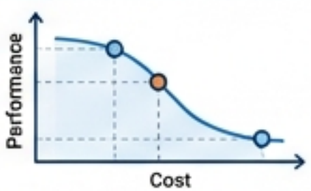
Cost/latency



Trace diffs



Pareto view



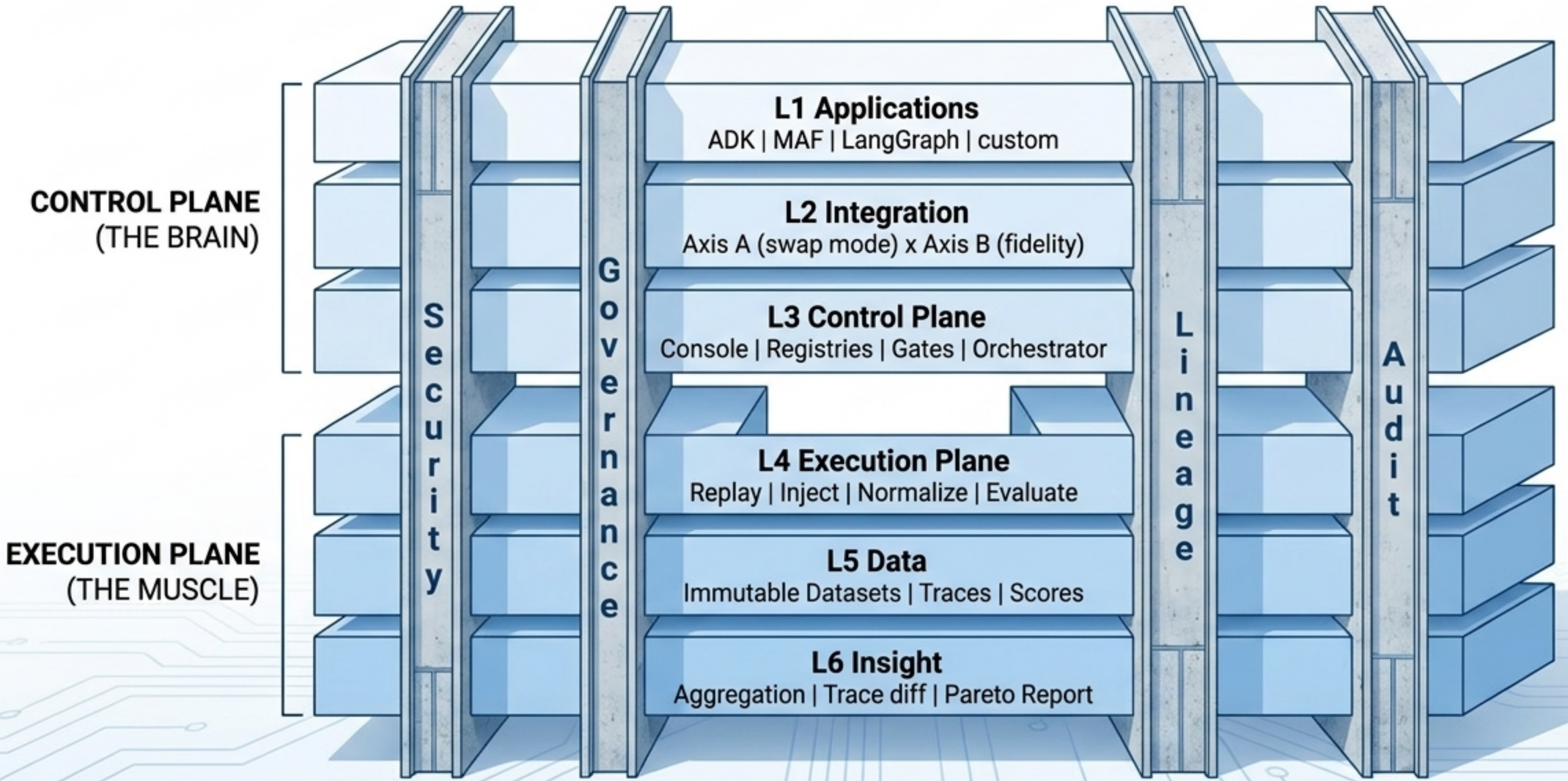
Trade-off statement

Trade-off: Model A offers 15% lower latency at 18% higher cost compared to Model B, with minor trace deviations in complex tasks. Consider based on priority.

What MIRA is NOT: It does not deploy to production, optimize prompts, touch live traffic, or rank models with a single arbitrary score. It produces **evidence**. **People decide.**

Gemini Notebook

A six-layer architecture separates governance from execution



Gemini Notebook

Automated discovery turns application code into a validated profile

The Scanner

grep anchors
AST patterns
read 5% slice

read **5%** slice

Draft Profile

```
{
  "Framework": "LangChain",   // src/main.py:45

  "Agents": [                 // src/agents.py:112
    "Summarizer",
    "Extractor"]
  ],
  "Tools": {
    "read-only": [
      "SearchAPI"]
    ],
    "idempotent": [
      "Calculator"]
    ],
    "stateful": [
      "DatabaseRead"]
  }
}                             // src/agents.py:112
```

Human Confirmation



Smoke Test

Invocation

Success

The Gates

- [x] Profile complete
- [x] Stateful tools have restore_snapshot hooks
- [x] Smoke test passes
- [x] Inventory matches trace

Validated Profile Ready

Gemini Notebook

Granular visibility depends entirely on the integration strategy

Axis A - Injection Mode	Columns: Axis B - Fidelity Level: full	near-full	decisions-only	black-box
adapter-intercept	✓	✓	✗ Adapter inherently sees more	✗ Adapter inherently sees more
config-seam	✓	✓	⚠ Risky	✓
gateway-route	⚠ Gateway translation enters the unit under test	⚠ Gateway translation enters the unit under test	✓	✗
variant-deployment	✓	✓	⚠ Risky	✓

MIRA evaluates the integration bundle (Model + Provider Adapter), not a pure model effect.

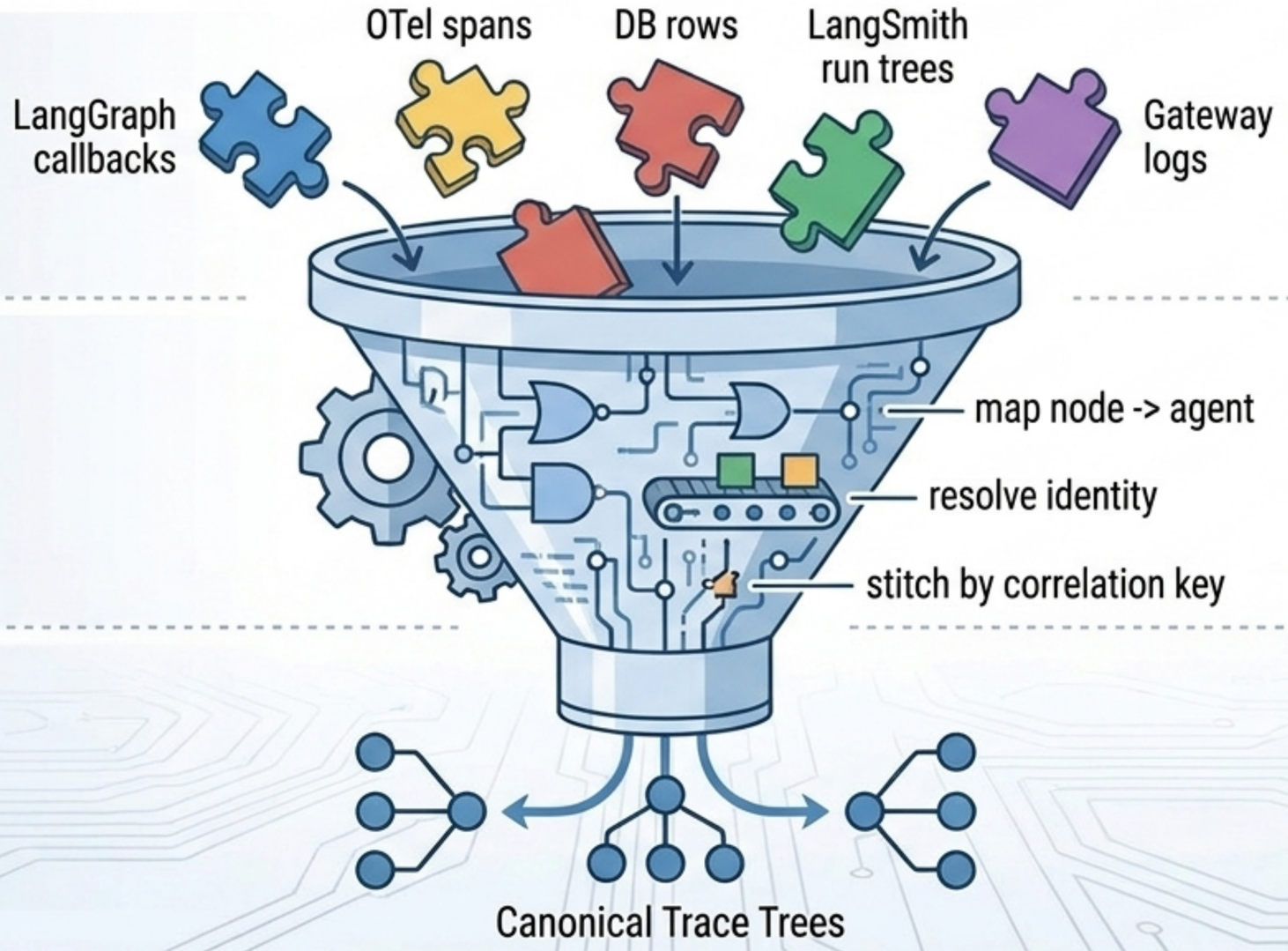
Gemini Notebook

A universal normalizer translates diverse telemetry into canonical traces

Chaos

Connectors & Mapping

Order

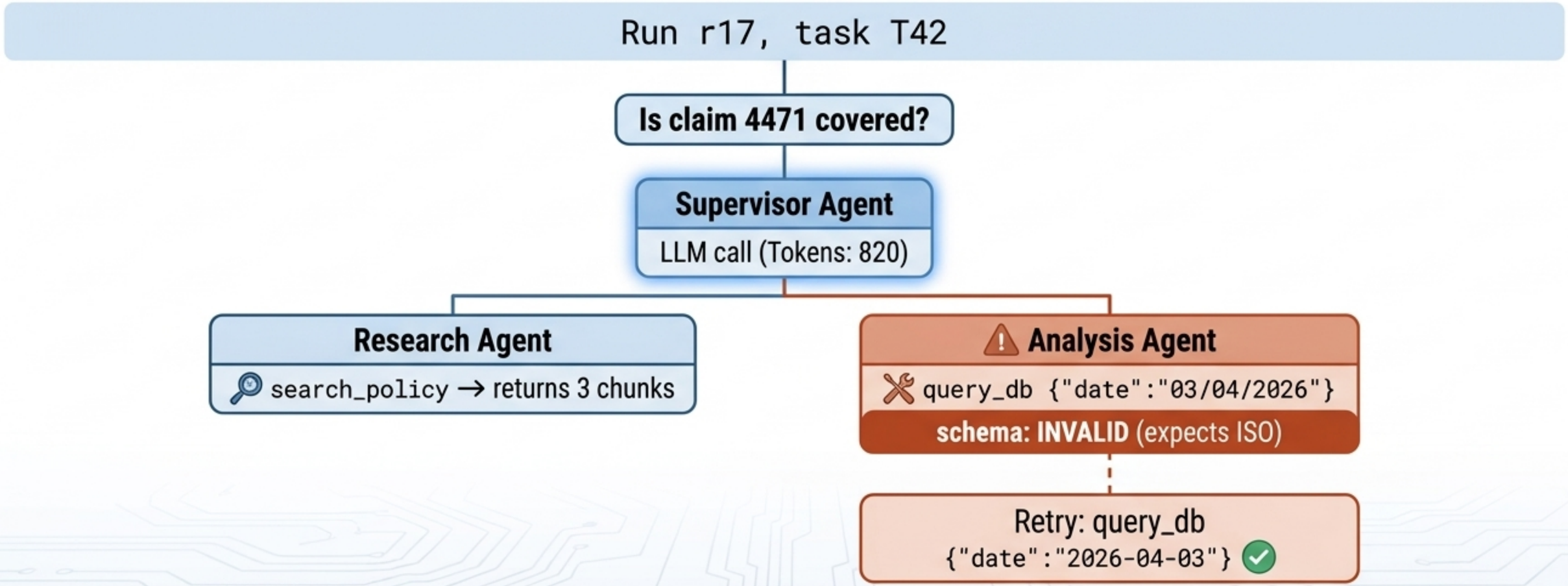


Gaps are strictly flagged, never guessed.

Unattributed spans lower the recorded coverage score.

Gemini Notebook

Canonical traces reveal the exact trajectory of every task

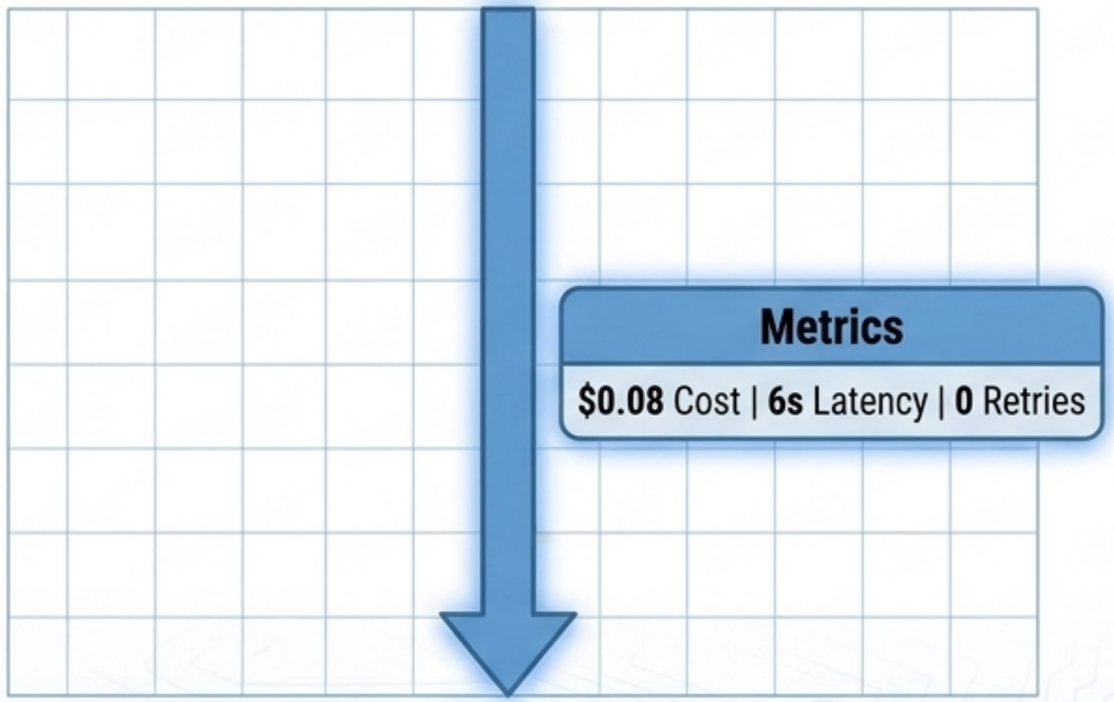


Independent of framework or code, evaluators read only this canonical trace.

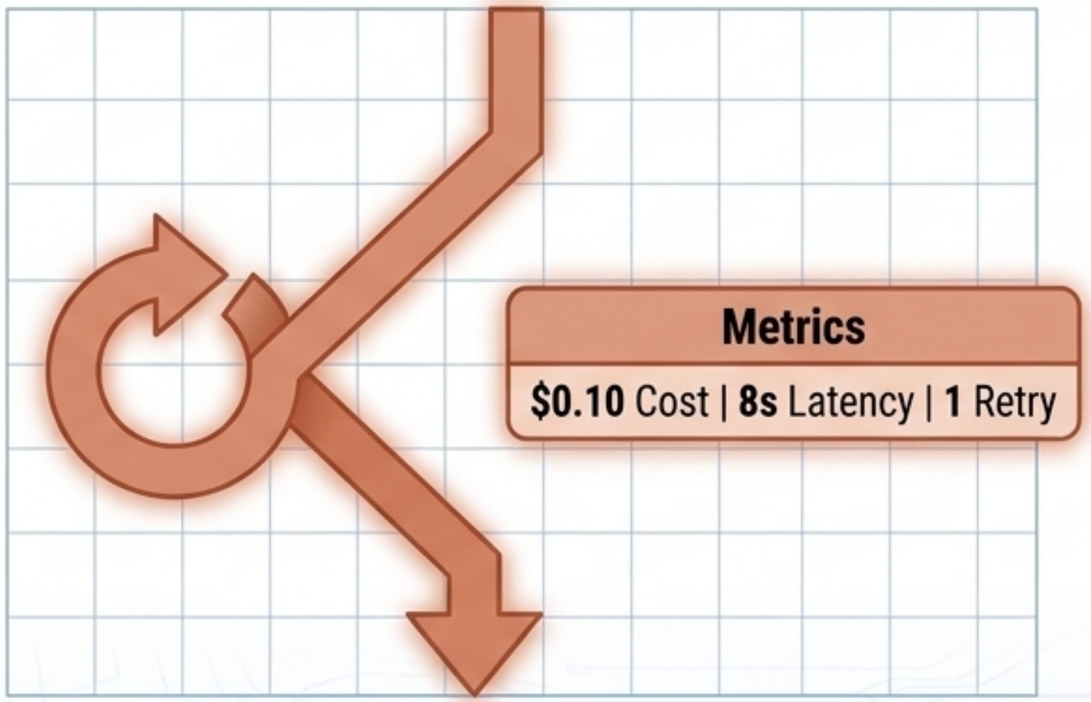
Gemini Notebook

Outcome success often hides the friction of recovery

Path A: First-Attempt Clean Success



Path B: Eventual Success with Retries

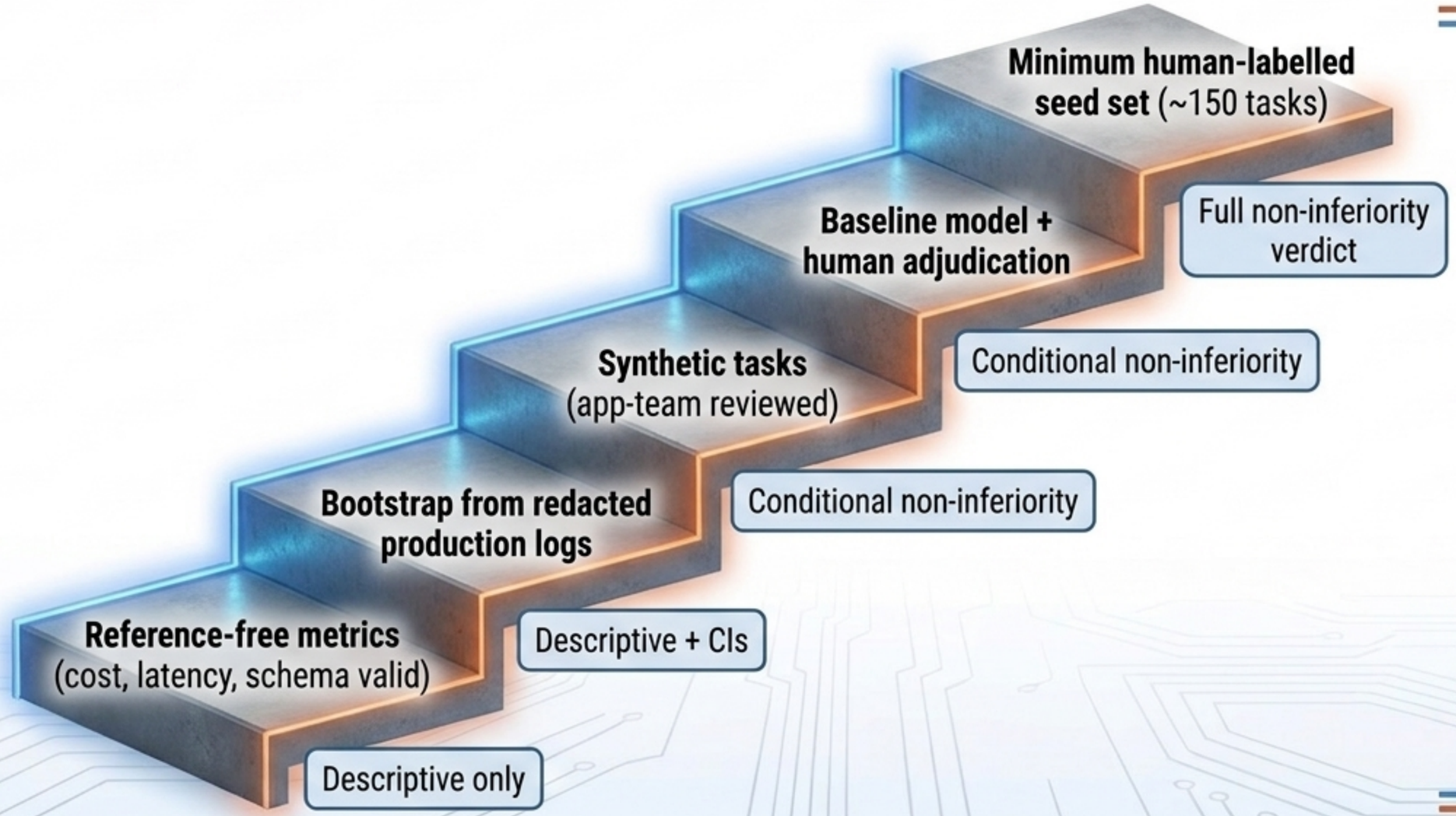


CPST = (Total cost of ALL runs, incl. failed) / (Number of successful runs)

Both paths achieve 'Outcome Success', but Path B fails 'Clean Success'. Retries heavily penalize Cost Per Successful Task.

Gemini Notebook

The strength of the final verdict scales with dataset quality



Lack of golden data does not block execution, it simply caps the verdict strength.

Gemini Notebook

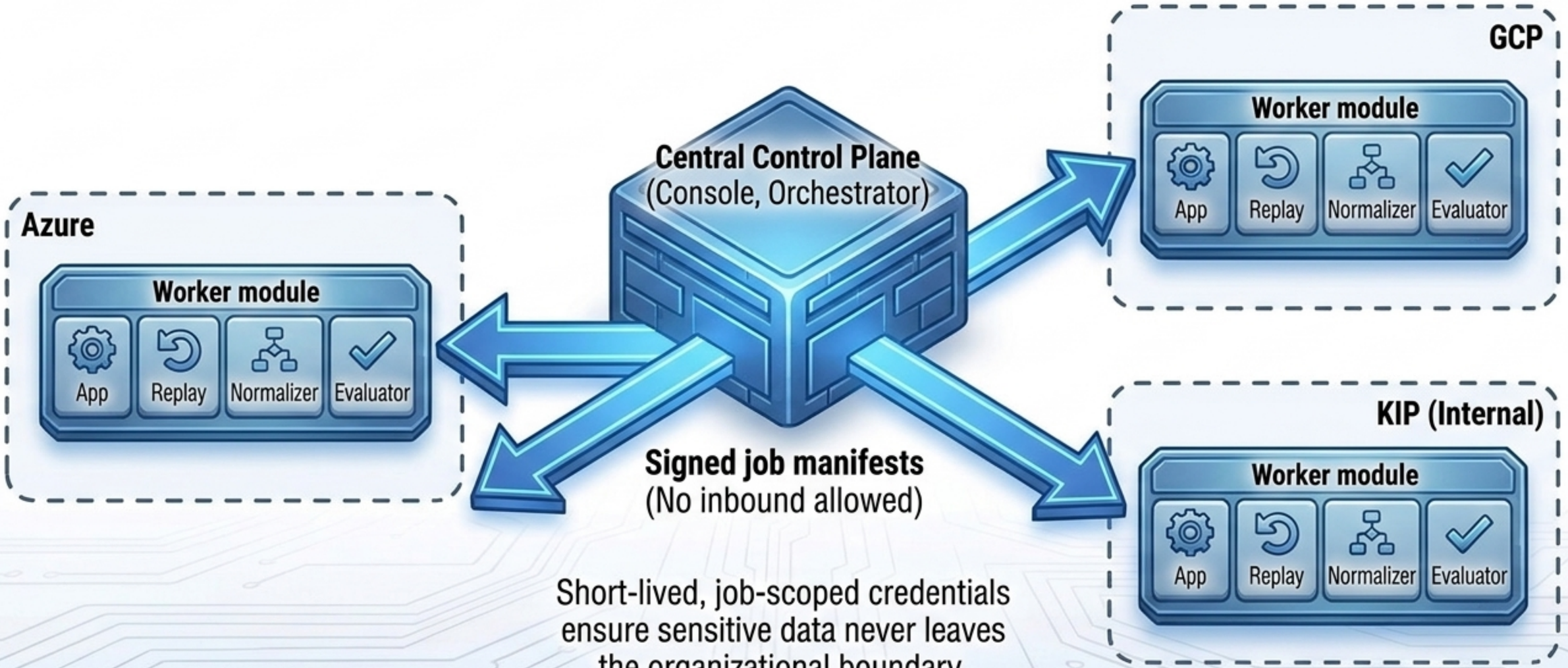
Experiments execute in a strictly controlled, immutable sandbox

Draft → Validated → Frozen → Running



Gemini Notebook

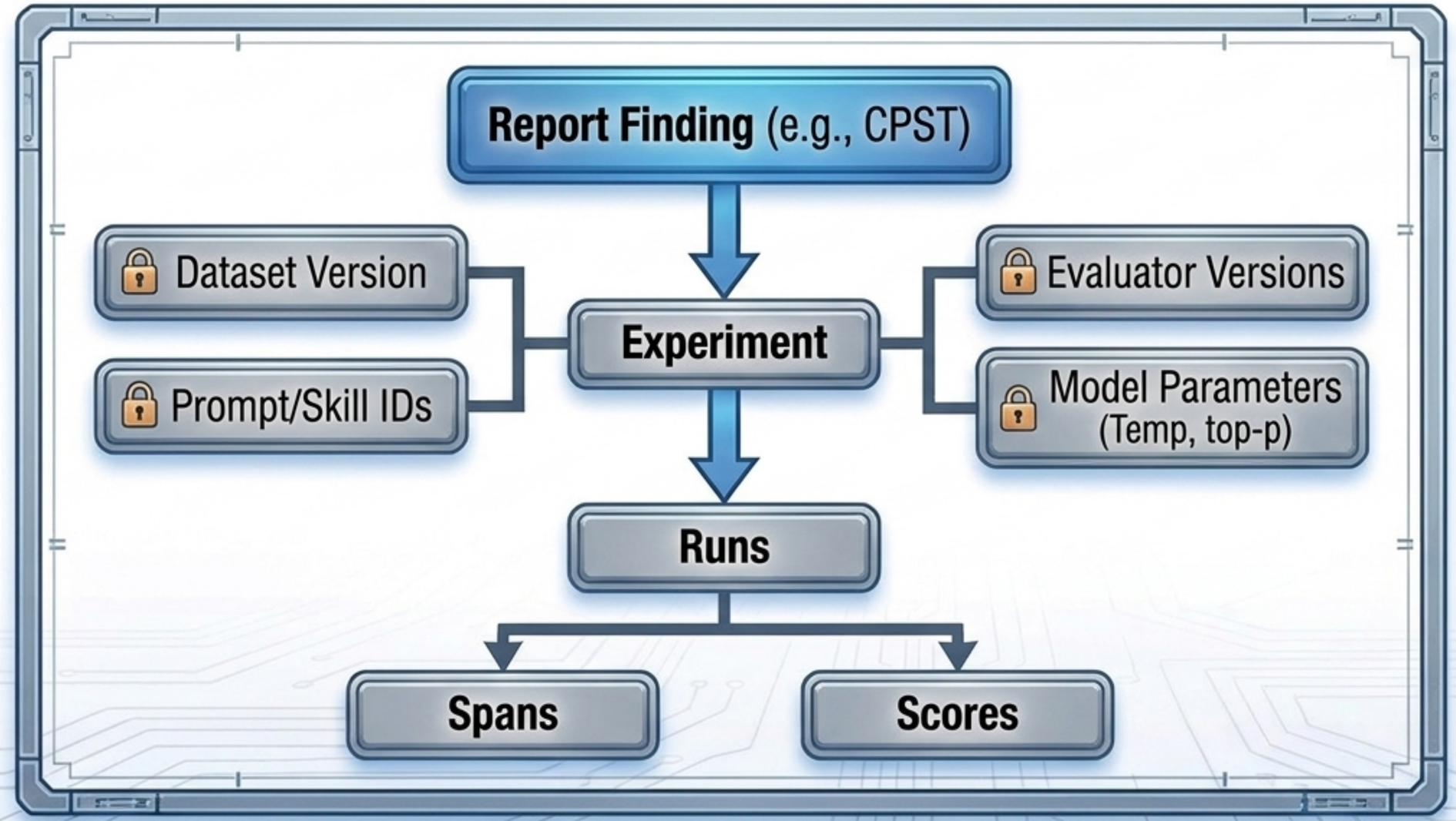
Remote execution workers move to your data boundaries



Short-lived, job-scoped credentials ensure sensitive data never leaves the organizational boundary.

Gemini Notebook

Every metric reported is anchored to immutable evidence

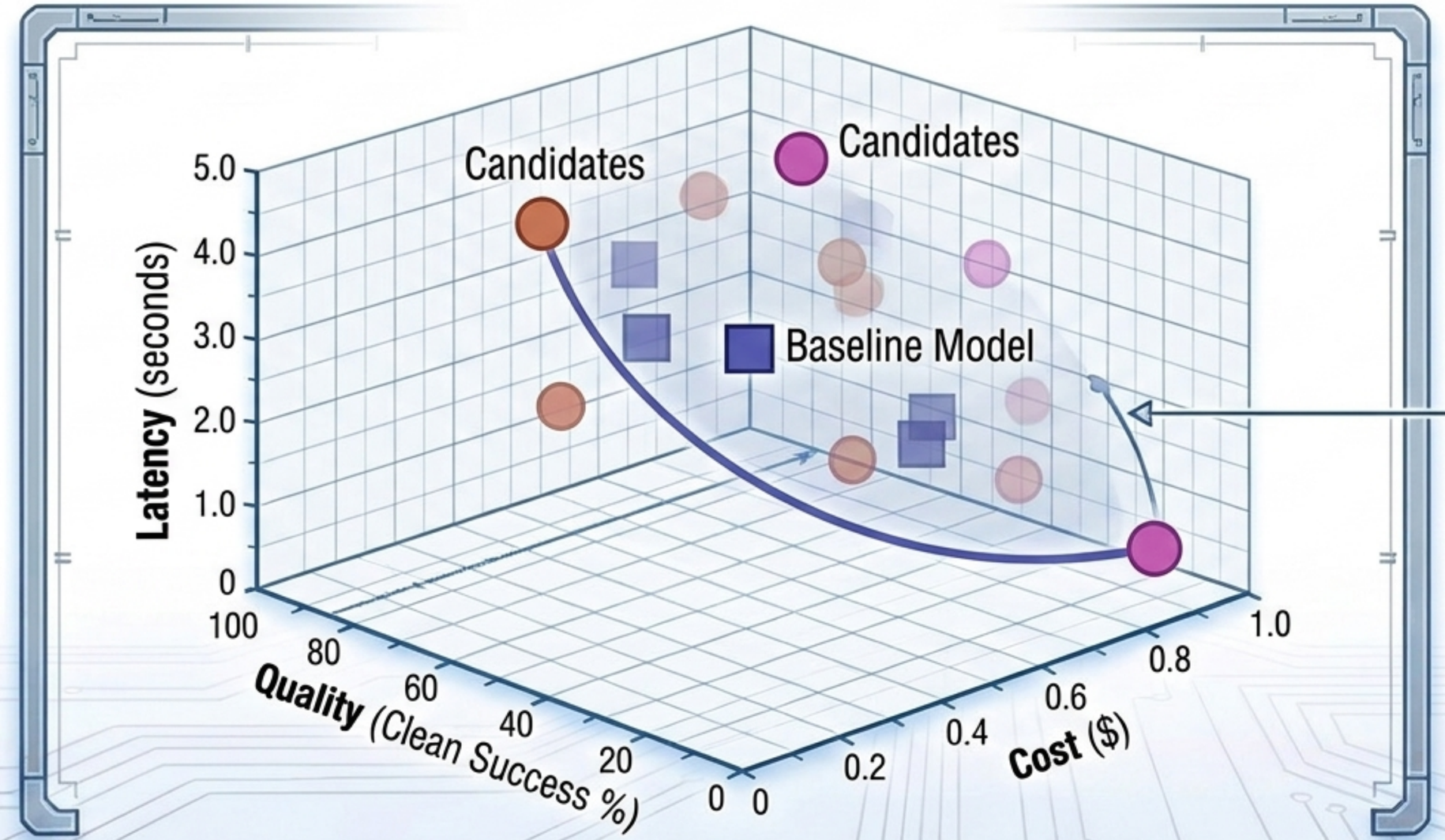


Changing a prompt or tool schema creates an entirely new experiment.

Results lacking a complete lineage record are automatically rejected.

Gemini Notebook

A multi-dimensional trade-off space replaces arbitrary rankings



MIRA does not rank models 1st, 2nd, and 3rd. It identifies the Pareto Front.

To prove non-inferiority, the candidate's lower confidence bound must exceed the strict margin (-δ) set before the run.

Gemini Notebook